Factors Influencing Personal Health Information Sharing for Digital Health Interventions Supporting Habit Formation: A Systematic Review

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Abstract: Digital health interventions show great promise in promoting long-term health behaviors; however, their effectiveness is contingent upon individuals' willingness to share personal health data. This willingness is essential, as data sharing enables interventions to be personalized and continuously refined to better suit individual needs, thus improving their relevance and effectiveness. This systematic review investigates the key factors influencing participation in health data sharing within habit-focused digital health research. Guided by the PRISMA methodology, 16 studies were analyzed, revealing 11 critical determinants—including trust in data recipients, privacy concerns, perceived risks, transparency, incentives, and user control. To synthesize these findings, a practical framework was developed, highlighting three core dimensions: personalization, trust and privacy, and incentives. Grounded in social exchange theory, the theory of reasoned action, and the theory of planned behavior, the framework offers actionable strategies for designing secure, user-centered data-sharing platforms that support sustainable health habit formation.

1 INTRODUCTION

Health behavior change represents a critical challenge in public health, with individuals seeking to modify behaviors ranging from dietary practices to exercise routines. In response, digital health interventions have emerged as promising tools, with the global digital health market projected to reach $659.8 billion by 2025 (Abernethy et al., 2022). However, for these interventions to achieve long-term success, they must move beyond temporary behavioral modifications to establish sustainable habits—automatically executed behaviors that, once formed, require minimal conscious effort (Wood et al., 2002; Lally and Gardner, 2013).

Digital recommender systems show particular promise in supporting this habit formation process by providing personalized, context-aware prompts that help users establish the consistent behavioral patterns necessary for habit development (Stark et al., 2023; Reinsch et al., 2024). Unlike generic interventions that produce only temporary effects (Beleigoli et al., 2019; Berry et al., 2021), these adaptive systems can tailor their recommendations to individual contexts, progressively building the cue-behavior associations fundamental to habit formation.

However, the effectiveness of personalized health recommender systems depends on fit-for-purpose PHI quality, as recommendation relevance typically improves with more accurate and consistent inputs (Heinrich et al., 2021). This data enables the algorithms to understand individual patterns, preferences, and contexts—essential elements for delivering timely, relevant interventions that can scaffold habit formation (Strotbaum et al., 2019; Reinsch et al., 2025b). Without sufficient personal data, these systems cannot provide the personalized support necessary for sustainable behavior change.

Yet a critical barrier emerges: despite the proliferation of health data generated through wearables and mobile applications, individuals remain reluctant to share their personal health information with digital health platforms and researchers (D3I, 2025; Strotbaum et al., 2019). This reluctance stems from various concerns including privacy risks, data misuse, and lack of transparency about data handling practices (Alam et al., 2024). This creates a fundamental tension: the very data needed to enable effective habit-supporting interventions is the data individuals are most hesitant to share.

Therefore, this systematic review examines the

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factors influencing individuals' willingness to share personal health information specifically for digital health interventions designed to support habit formation. The focus is not on habitual data-sharing behaviors themselves, but rather on understanding what motivates or inhibits individuals from providing the data necessary for habit-supporting digital health systems. By synthesizing existing evidence and grounding findings in established behavioral theories, this study aims to provide actionable insights for designing user-centered, ethically sound data-sharing platforms.

The specific research questions guiding this review are:

- **RQ1**: What factors influence individuals' willingness to share personal health information for digital health interventions aimed at habit formation?
- **RQ2**: How do established behavioral theories explain these factors and their interrelationships?

2 THEORETICAL BACKGROUND

2.1 The Importance of Data Sharing for Habit-Based Interventions

A habit is defined as an automatically executed behavior consistently triggered by a specific context, formed through repeated association between contextual cues and a specific action (Neal et al., 2006; Lally and Gardner, 2013). Thus, a habit fundamentally consists of two components: the repetitive behavior itself and the context in which it reliably occurs (Neal et al., 2006). Once fully developed, habits require minimal conscious effort, enhancing their potential to sustain long-term health improvements by reducing cognitive load and fostering behavioral automaticity (Wood et al., 2002). While numerous digital health interventions have demonstrated effectiveness in promoting immediate behavioral change—such as increased physical activity or dietary adjustments—they often struggle to maintain these effects long-term due to their reliance on ongoing user engagement and external cues such as notifications or reminders (Beleigoli et al., 2019; Berry et al., 2021). In contrast, established health habits inherently facilitate persistent, automatic behaviors independent of continuous external prompting, thereby potentially offering sustained health benefits over prolonged periods. However, habit formation itself is complex and often requires an initial external stimulus or motivational "push" to encourage consistent behavior repetition, a role effectively fulfilled by digital recommender systems (Stark et al., 2023; Reinsch et al., 2024). Digital recommender systems offer personalized prompts and recommendations that support users in establishing mental associations necessary to embed new health behaviors into their daily routines, thereby facilitating habit formation. These systems rely on high-quality, personalized user data to train algorithms that effectively adapt recommendations to individual contexts. Despite the rapid growth in personal health data generation—particularly through widespread use of wearables and health-tracking applications—much of this valuable data remains proprietary, stored exclusively by single providers, and largely inaccessible to researchers or healthcare professionals (Strotbaum et al., 2019). In this context, data sharing—defined as granting second- or third-party access to personal health data or personal health information (PHI) is crucial. Second-party access refers to sharing data with the primary service provider or platform delivering the digital health intervention, while third-party access involves sharing with external entities such as researchers, academic institutions, or other organizations beyond the primary service provider. For researchers, access to shared PHI represents a significant opportunity to refine and personalize digital interventions, enhance clinical trials, and address pivotal healthcare questions concerning disease prevention, diagnosis, and therapy (Strotbaum et al., 2019). For individuals, this practice can offer immediate benefits such as tailored disease prevention programs, alongside broader long-term health improvements and advancements in treating chronic or hereditary diseases (Strotbaum et al., 2019; Reinsch et al., 2025a). Yet, despite these promising opportunities, data sharing remains constrained by several barriers, including technical limitations, lack of interoperability, data monopolies, and privacy concerns (D3I, 2025; Strotbaum et al., 2019). Therefore, fostering individuals' willingness to participate in health data sharing emerges as a key priority for advancing habit-focused digital health interventions, ultimately supporting sustained improvements in public health outcomes.

2.2 Theoretical Perspectives on Health Data Sharing Behavior

To understand why individuals choose to share their PHI, it is essential to examine the underlying motivational factors that influence their behavior. Social Exchange Theory (SET), visualized in Figure 1, offers a valuable lens for this analysis by proposing that individuals evaluate the perceived balance of benefits and costs when deciding whether to engage in a specific behavior (Thibaut and Kelley, 1959). The *social exchange* outcome in Figure 1 represents the individual's decision to share or withhold PHI based on their evaluation of benefits versus costs, mediated by trust in the data recipient (Dillman et al., 2014). In this context, benefits, understood as either external (e.g., monetary or non-monetary incentives) or internal (e.g., personal satisfaction or emotional gratification), are weighed against potential risks such as privacy loss or data misuse (Sammut et al., 2021; Toepoel, 2012). Research indicates that these motivations can stem from both egoistic and altruistic impulses. Prior research has highlighted that personal motivations for survey participation can be classified in several distinct ways, and that while appeals to societal benefit may enhance participation, egoistic incentives such as personal benefits generally prove more effective when combined with altruistic appeals (Kropf and Blair, 2005; Singer and Ye, 2013). SET has since been expanded to include the concept of trust, defined as the participant's belief that the benefits of participation will outweigh the costs and that the data recipient will adhere to promised terms of use (Dillman et al., 2014; Sammut et al., 2021). Importantly, SET acknowledges that individuals do not always act based on deliberate, rational calculations; rather, decisions are often made quickly, influenced by emotional and contextual cues (Dillman et al., 2014). Thus, SET provides a comprehensive theoretical foundation for understanding PHI sharing behavior by highlighting the importance of perceived benefits, trust, and the interplay of altruistic and egoistic motivations.

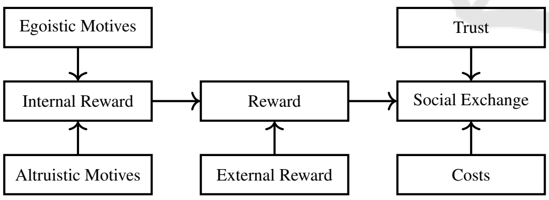

Figure 1: Social Exchange Theory Model.

In addition to SET, the Theory of Reasoned Action (TRA) and the Theory of Planned Behavior (TPB), visualized in Figure 2, offer further insight into the psychological mechanisms underlying individuals' decisions to share PHI (Fishbein and Ajzen, 1975; Ajzen, 1991). Both theories emphasize the central role of behavioral intention as the most proximal predictor of actual behavior and have been widely applied in health-related contexts such as blood donation and healthcare service utilization (Glanz et al., 2015). According to TRA, behavioral intention is shaped by two primary determinants: an individual's attitude toward the behavior and the subjective norms surrounding it. Attitude refers to the person's overall evaluation of the behavior based on anticipated outcomes, while subjective norms reflect the perceived social pressure to engage, or not engage, in the behavior, as well as the individual's motivation to comply with the expectations of others. However, TRA assumes that behavior is under volitional control, which may not always be the case in real-world scenarios. To address this limitation, TPB extends TRA by incorporating perceived behavioral control as a third determinant (Ajzen, 1991). This construct captures an individual's beliefs about the presence of facilitating or inhibiting factors, such as time constraints, digital literacy, or privacy concerns, and the perceived difficulty or ease of performing the behavior (Glanz et al., 2015). Importantly, perceived behavioral control is a belief construct and may diverge from actual control; objective constraints can therefore limit whether intentions result in PHI sharing even when perceived control is high. In the context of PHI sharing, TPB provides a useful framework to explain how attitudes (e.g., trust in institutions), subjective norms (e.g., expectations of contributing to research), and perceived behavioral control (e.g., confidence in managing consent and data sharing settings) jointly influence individuals' willingness to disclose personal health data.

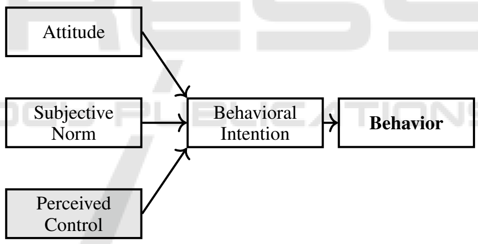

Figure 2: Theory of Reasoned Action and Theory of Planned Behavior Model.

Here, "PHI sharing" refers to contributing habit-related PHI for cross-individual aggregation to train and evaluate habit recommender systems. This differs from disclosure for personalization within a closed system, which is outside our scope. This paper systematically reviews the existing literature on the factors and incentives that influence participation in health data sharing within the context of digital health habit research. Through the application of a structured review methodology, the study identifies and categorizes key determinants of individuals' willingness to share PHI. These factors are then examined in relation to established behavioral theories, includ-ing SET, the TRA, and the TPB, to provide a theoretical foundation for understanding PHI sharing behavior. Moreover, the review explores why certain factors are cited more frequently across studies, offering insights

into the underlying mechanisms that may enhance or inhibit data-sharing behavior. Based on these insights, a practical framework is developed to systematize the identified factors and generate actionable implications for the design of data-sharing platforms in scientific research. The central research question guiding this review is as follows:
What factors influence participation in health data sharing in digital health habit research, and which theoretical concepts underlie them?

3 METHOD

This study was conducted as a systematic literature review, aiming to provide a comprehensive overview of the factors and incentives influencing participation in health data sharing. The PRISMA guideline was used to ensure methodological rigor (Page et al., 2021; Moher et al., 2009). To ensure reproducibility, the online tool CADIMA was utilized, developed for systematic literature reviews (Kohl et al., 2018). Furthermore, the tool offers the ability to summarize steps of the systematic review in various forms of presentation formats. Graphics and tables generated by CADIMA were labeled accordingly. The scope defined by the research question is reflected in the search process. Information sources were selected based on their coverage of the targeted research domain digital health habit research, as well as the availability of suitable results from an initial scoping search. Due to the interdisciplinary nature of the research field, journals as well as databases were chosen as information sources to ensure a broader coverage of relevant studies. This approach enabled a more comprehensive inclusion of available research, incorporating related fields such as behavioral science. While medical databases such as PubMed were considered, we selected Scopus, Web of Science, ProQuest, and JMIR Human Factors based on their comprehensive coverage of interdisciplinary research spanning behavioral science, digital health, and information systems—domains central to understanding health data sharing behavior in digital health contexts. These databases collectively index journals from medical informatics, human-computer interaction, and behavioral psychology, providing broader coverage than purely medical databases alone. Key concepts of the research question formed the basis for search terms, while additional terms were identified through thesaurus searches for synonyms. During the scoping search recurring concepts that were either unrelated to the research field or disconnected from the research question were identified. These concepts were then incorporated as exclusion terms on the abstract level to refine the search for the systematic review. The terms for inclusion and exclusion are shown in Table 1.

Table 1: Inclusion and exclusion search terms used to identify relevant literature.

Inclusion Terms	Exclusion Terms
incentiv*	covid-19
motivat*	pandemic
rewar*	blockchain
data sharing	machine learning
shar*	data sensing
data donation	research data management
donat*	
data contribution	
disclos*	
personal data	
health data	
digital health	
*health	
habit*	
habit research	
routine behavio*	

The search terms were structured into comprehensive search strings by combining synonyms within the same conceptual category (e.g., data sharing, data donation) using the Boolean operator OR, while distinct conceptual categories (e.g., data sharing and digital health) were connected using AND. One example of a resulting search string used in the Scopus database is as follows:

((“data sharing” OR “data donation” OR “data contribution”) AND (“digital health” OR health) AND (habit OR “routine behavior”))

To refine the search and reduce irrelevant results, exclusion terms identified during the scoping phase (see Table 1) were incorporated using NOT operators. Furthermore, the search was limited by applying filters to include only peer-reviewed academic publications written in English, published between 2005 and 2025, and categorized as articles, reviews, or conference papers. This ensured the inclusion of high-quality, relevant sources aligned with the scope of this review. The study selection process was based on the four-phase flow diagram as proposed by the PRISMA guideline and is shown in Figure 3 (Page et al., 2021). To assess the eligibility of articles for inclusion in the systematic review, study eligibility criteria were defined using the PICOS approach (Liberati et al., 2009). Since this review aims to identify the factors influencing the outcome participation in health data sharing, any study examining the general population aged 18 and above was considered eligible. Additionally, studies utilizing an incentive to influence participation in health data sharing, had to use a non-monetary incentive as an intervention. The criteria for comparator and study design were intentionally not limited in this review so as not to exclude relevant studies in research based on these factors. The initial search across all databases yielded 1,472 articles. After removing 109 duplicates, 1,363 articles remained for title/abstract screening. Of these, 1,332 were excluded based on title/abstract review, leaving 31 full-text articles for assessment. Following full-text review, 15 articles were excluded for the following reasons: studies targeting very specific populations with small sample sizes (n=3) and studies not reporting on willingness to share health data (n=12). This resulted in 16 articles included in the final synthesis.

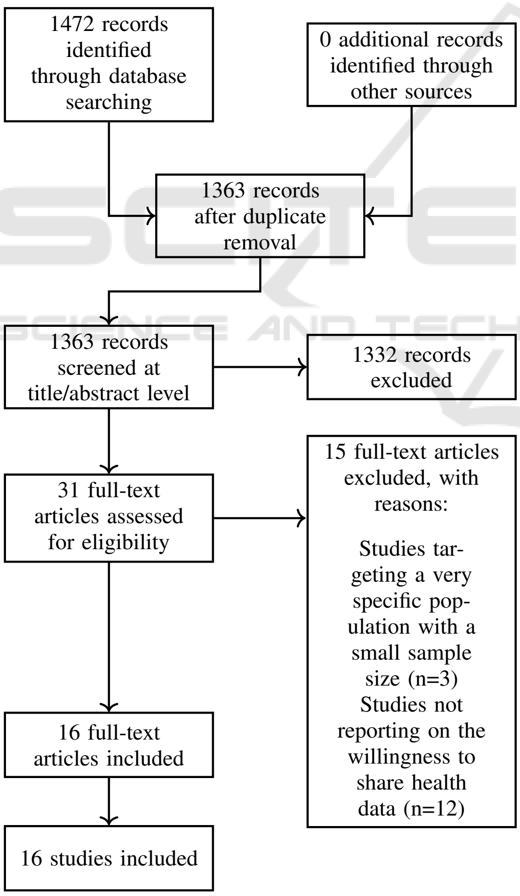

Figure 3: PRISMA Flow Diagram.

The analysis included studies conducted across Europe (9), North America (3), South America, and Asia (3), with one systematic review referencing unspecified countries. Most studies (12/16) explored attitudes and factors influencing individuals' willingness to share personal health information (PHI), while others examined privacy protections (2), message framing, and nudging interventions (2). Methodologically, most used quantitative approaches (12/16), including surveys and experiments (some as discrete choice), while others employed qualitative or mixed methods. Thirteen studies conducted primary research; three were literature reviews. The Gioia methodology (Gioia et al., 2013) was applied for qualitative coding of identified factors. This involved three stages: (1) initial coding of factors as mentioned in each study using in-vivo codes, (2) grouping similar concepts into first-order categories through collaborative discussion, and (3) abstracting these into theoretical dimensions linked to behavioral theories. For example, factors such as "data security," "privacy protection," and "risk of misuse" were grouped under the broader category of "Privacy, Security and Risk."

4 RESULTS

Table 2 summarizes the 16 included studies and indicates which of the 11 identified factors each study addressed. The factors are defined in Table 3 and their relative frequency across studies is visualized in Figure 4.

One of the most frequently cited was the data recipient, with participants showing greater trust in public or research institutions over private companies (Vilaza et al., 2021). Sharing PHI for research was generally preferred to sharing for commercial purposes (Alam et al., 2024; Biasiotto et al., 2023). The closely related factors of privacy, risk, and security also emerged as critical. Privacy referred to ensuring data confidentiality throughout collection and processing, while security encompassed systems to uphold this privacy (Benevento et al., 2023). Risk involved fears of data misuse, such as unauthorized access, discriminatory practices, or profit-making with-out consent (Alam et al., 2024; Lee et al., 2022; Ziefle and Calero Valdez, 2018). Additionally, benefits and incentives were found to positively impact data-sharing decisions. Financial incentives were strong motivators (Fernandes and Pereira, 2021; Ziefle and Calero Valdez, 2018), while intrinsic motivations such as contributing to research and altruism also played a significant role (Karampela et al., 2019). The importance of transparency and self-management was highlighted in studies emphasizing participants' desire to be informed about how their data would be used, and to retain control over sharing decisions (Biasiotto et al., 2023; Johansson et al.,

Table 2: Mapping of identified factors across included studies.

Study	Method	Data Recipient	Privacy, Security, Risk	Benefit & Incentive	Information & Transparency	Data type	Policies & Consent	Purpose of Use	Socio-demographic Variables	Data Collector	Habits	Message Frame
(Wang, 2024)	empirical											X
(Benevento et al., 2023)	literature review		X			X		X	X			
(Zeller et al., 2024)	empirical		X		X							
(Lee et al., 2022)	empirical	X	X	X	X							
(Guo et al., 2022)	empirical			X								
(Fernandes and Pereira, 2021)	empirical		X	X							X	
(Vilaza et al., 2021)	empirical	X	X	X	X	X						
(Alam et al., 2024)	literature review	X	X	X		X						
(Biasiotto et al., 2023)	empirical	X			X		X	X		X		
(Gupta et al., 2023)	empirical				X		X	X	X			
(Schöning et al., 2019)	empirical		X				X					
(Karampela et al., 2019)	empirical	X		X					X			
(Johansson et al., 2021)	empirical	X			X		X	X		X		
(Ziefle and Calero Valdez, 2018)	empirical	X	X	X		X						
(Naeem et al., 2022)	literature review	X	X	X		X			X			
(Muller et al., 2022)	empirical	X					X					

2021). Relatedly, the purpose of use was influential: PHI use for research was broadly accepted, whereas marketing and advertising purposes elicited skepticism (Gupta et al., 2023; Johansson et al., 2021). Willingness also varied based on the type of data shared. For instance, location data prompted discomfort (Vilaza et al., 2021), while physical activity data was more readily shared. DNA data had low sharing rates (Benevento et al., 2023), except among individuals with illnesses like cancer, who were more inclined to share in hopes of contributing to improved treatments (Alam et al., 2024). The role of policies and consent mechanisms was significant, with consent frameworks, particularly those enabling data deletion and condition-based sharing, positively influencing willingness (Gupta et al., 2023; Muller et al., 2022). Socio-demographic factors such as age, education, and occupation were also relevant: older individuals, pensioners, and those with lower educational attainment were less likely to share PHI, likely due to increased concerns over data misuse and diminished digital literacy (Benevento et al., 2023; Gupta et al., 2023; Karampela et al., 2019). The data collector's identity influenced trust as well, with academic institutions and healthcare providers seen as more trustworthy than technology firms (Biasiotto et al., 2023; Johansson et al., 2021). In terms of behavioral models, prior research found that attitudes, perceived behavioral control, and social benefits (from the TRA and SET frameworks) all positively influenced the sharing of electronic health records (Guo et al., 2022). Furthermore, framing effects played a role: presenting PHI sharing as "donating" increased willingness when a strong social cause was perceived (Wang, 2024). Lastly, prior disclosure behavior was a strong predictor, suggesting that willingness to share PHI may become habitual over time (Fernandes and Pereira, 2021).

The 16 included studies exhibited considerable heterogeneity in methodological approach and rigor. Sample sizes ranged dramatically from 25 participants in qualitative studies to 36,268 in large-scale surveys, with a median of approximately 450 participants. This variation raises important considerations about the relative weight of findings. Three studies were literature reviews rather than primary research, potentially introducing secondary interpretation bias. The geographic distribution revealed a Western bias, with 75% of studies (12/16) conducted in Europe or North America, limiting the generalizability of findings to other cultural contexts where attitudes toward data privacy may differ significantly.

Table 3: Identified factors influencing individuals' willingness to share personal health information (PHI).

Factor	Description
Data recipient	The stakeholder that receives and utilizes the shared health data.
Privacy, Security, Risk	Assurance that health data is treated confidentially using secure systems and processes, minimizing the risks of data compromise.
Benefit/ Compensation/ Incentive	benefits individuals expect from sharing their health data.
Information, Transparency and Self-Management	Transparency of the data-sharing process and the ability to make decisions about one's own data.
Type of Data/Information	The nature of the underlying medical data and technical data characteristics.
Policies and Consent	Data privacy policies explaining how health data is collected, used, and shared. Consent to privacy policies.
Purpose of Use	The use of the collected health data.
Socio-demographic Variables	The characteristics that describe specific population groups.
Data Collector	The entity that requests information from individuals and forwards it to the data recipient.
Habits	Data-disclosure habits of individuals.
Message Frame	The framing of the message of an appeal for the sharing of data.

5 DISCUSSION

The analysis identified 11 factors influencing PHI sharing. Grounding these factors in established psychological theories not only strengthens academic rigor but also clarifies the interconnections among them. This theoretical grounding provides a comprehensive understanding of PHI sharing behaviors, ultimately enabling the derivation of practical implications. SET suggests that individuals evaluate the balance between costs and benefits when deciding whether to perform a specific behavior. Accordingly, the identified factors, privacy, security and risk, as well as benefit, compensation, and incentive, directly relate to these cost-reward assessments. Privacy, for instance, involves evaluating whether the potential risks of sharing PHI, such as data misuse or identity disclosure, outweigh the anticipated benefits (Fernandes and Pereira, 2021). High perceived privacy risks generally discourage PHI sharing, highlighting the crucial role privacy plays in influencing health data-sharing decisions. Conversely, tangible benefits or incentives positively influence PHI sharing, provided they sufficiently counterbalance or exceed the perceived privacy risks (Fernandes and Pereira, 2021; Schöning et al., 2019). The purpose of use is another factor explainable through SET. Studies indicate that individuals are significantly more willing to share PHI for research purposes compared to marketing or promotional objectives, suggesting that contributing to research provides a perceived social or personal reward that compensates for associated costs (Gupta et al., 2023). However, given that this factor was mentioned in only four studies, financial incentives may generally be more effective in motivating PHI sharing than altruistic appeals alone. The TRA posits that behavioral intentions are influenced by an individual's attitude toward the behavior and subjective norms, making it particularly suitable for examining factors related to personal attitudes and social dynamics. Socio-demographic variables, such as age, education, and occupational status, shape attitudes toward health data sharing; for instance, older individuals tend to exhibit greater privacy concerns and reluctance in sharing PHI (Gupta et al., 2023; Karampela et al., 2019). However, the limited coverage of these factors, addressed in only four studies, suggests inconsistencies within demographic groups. Indeed, research highlights mixed findings regarding socio-economic status: some studies report lower willingness to share PHI among lower socio-economic groups, while others show that low-income individuals are receptive if tangible benefits are offered (Benevento et al., 2023). Policies and consent mechanisms similarly align with TRA's determinant of attitude; clear and transparent consent processes positively influence willingness to share PHI (Gupta et al., 2023; Muller et al., 2022). Despite its importance, this factor appeared in just five studies, possibly due to individuals' frequent exposure to lengthy, non-engaging privacy policies, fostering a habit of automatic self-disclosure (Fernandes and Pereira, 2021), exacerbated by user-unfriendly presentation formats (Zeller et al., 2024). Furthermore, message framing aligns with

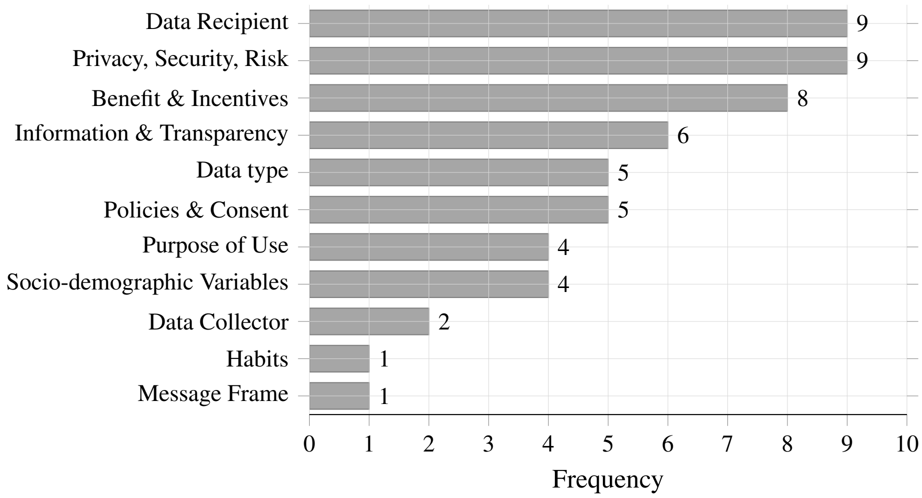

Figure 4: Identified Factors for Personal Health Information Sharing.

TRA's determinant of subjective norms. Framing PHI sharing as "donating" enhances disclosure when perceived social benefits are high, appealing to individuals' desire to conform to social expectations (Wang, 2024). The TPB extends TRA by introducing perceived behavioral control. Research has identified habits as predictors of data disclosure behavior; habits, being automatic actions formed through repeated behavior, require minimal conscious effort (Fernandes and Pereira, 2021; Verplanken and Wood, 2006). Applied to PHI sharing, repeated data disclosures can reinforce habitual disclosure behavior, thus reducing perceived barriers over time (Fernandes and Pereira, 2021). Furthermore, the factors data recipient and data collector influence perceived behavioral control, as individuals' willingness to share PHI strongly depends on trust in the entity handling their data. Individuals typically perceive greater control when sharing their health data with trusted entities, such as research institutions or healthcare providers, compared to commercial organizations (Biasiotto et al., 2023; Johansson et al., 2021). The data recipient emerged as particularly significant, whereas the data collector appeared less important, suggesting that individuals are more concerned with who ultimately uses their data rather than who initially collects it. Additionally, information, transparency, and self-management enhance perceived behavioral control; when individuals clearly understand with whom their PHI is shared and have mechanisms to monitor or manage data sharing, their sense of control increases (Biasiotto et al., 2023; Johansson et al., 2021). Lastly, the type of data shared is also related to perceived behavioral control. Sharing sensitive data, such as genetic or DNA information, is often associated with greater perceived barriers, reflecting uncertainty and discomfort due to unfamiliarity with potential consequences. Low willingness to share DNA data has been observed, reflecting individuals' selective attitudes toward different types of information (Middleton et al., 2020).

6 PRACTICAL IMPLICATIONS

To translate the identified factors into actionable strategies, this study proposes a framework aimed at enhancing participation in health data sharing on research-oriented platforms. The framework presented in Figure 5 synthesizes the 11 identified factors into three actionable branches, derived through thematic clustering of related concepts.

The first branch, 'Personalization and User Experience', combines factors related to interface design and user segmentation, including socio-demographic variables and habits. This branch emphasizes the need for a streamlined and user-friendly interface, including the implementation of default settings and user segmentation based on socio-demographic variables or platform roles, to reduce barriers and reinforce habitual disclosure behavior (Fernandes and Pereira, 2021; National Institutes of Health, 2010).

The second branch, 'Trust and Privacy', integrates factors concerning data security, transparency, and institutional credibility (data recipient, privacy/security/risk, information/transparency). This branch focuses on fostering trust by ensuring transparent data practices, robust security standards, and granular consent mechanisms, while also promoting the visibility and credibility of research institutions involved (Biasiotto et al., 2023; Gupta et al., 2023; Johansson et al., 2021; Lee et al., 2022).

The third branch, 'Compensation and Incen-

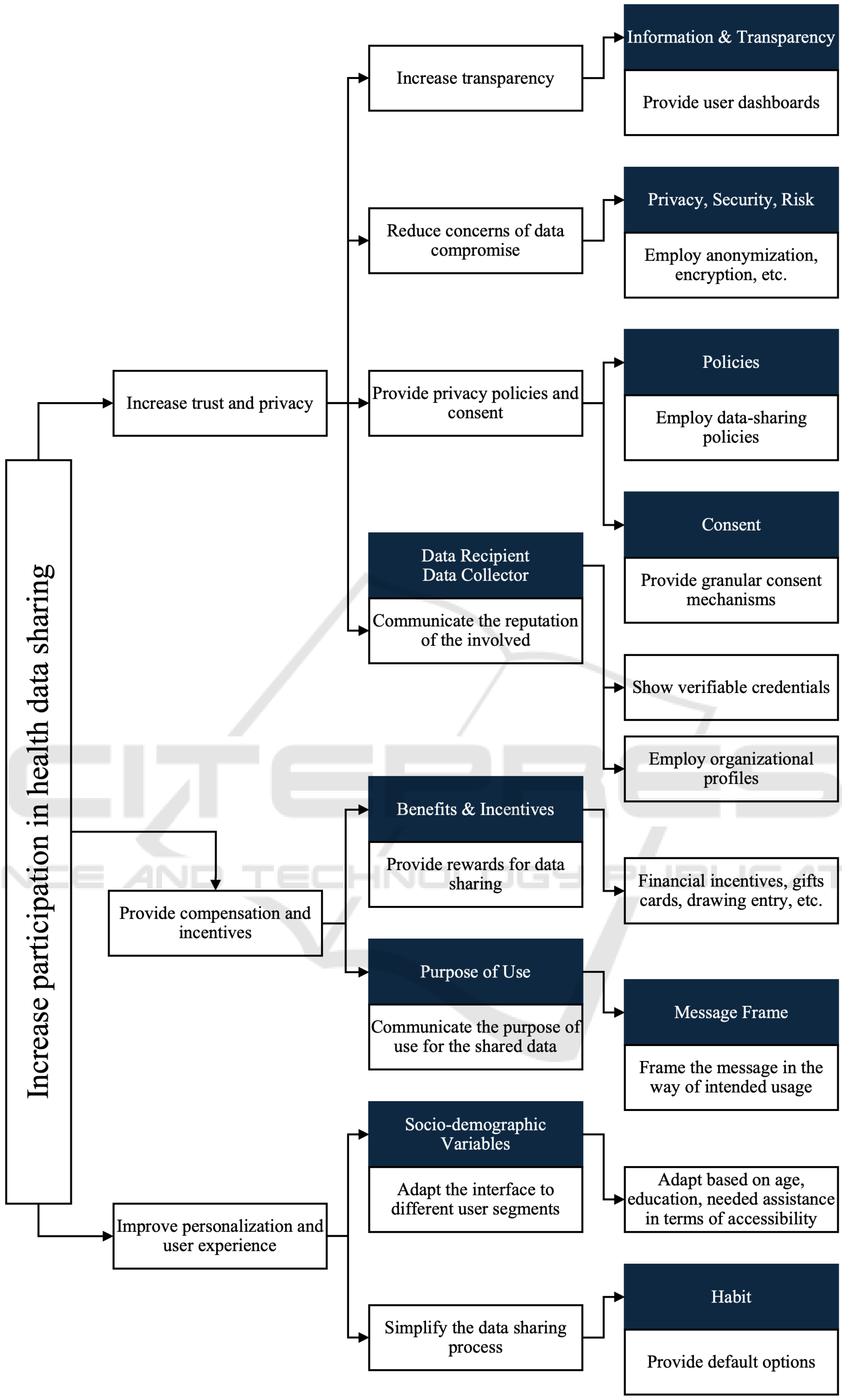

Figure 5: Framework for Increasing Personal Health Information Sharing.

tives', addresses motivational factors including benefits, message framing, and purpose of use. This branch recommends tangible incentives and effective communication of the research purpose using tailored message framing to underscore the societal impact of data sharing (Karampela et al., 2019; Naeem et al., 2022; Wang, 2024).

This framework is intended for three primary audiences: (1) researchers developing data collection protocols for habit-focused digital health studies, (2) platform designers creating user interfaces for health data sharing systems, and (3) policymakers establishing guidelines for ethical health data collection. The framework's modular structure allows stakeholders to focus on specific branches while maintaining awareness of their interconnections. Collectively, this framework provides a theoretically grounded and practically relevant approach to enhancing PHI sharing in the context of scientific research. Serving as both a foundation for researchers developing new interventions to enhance PHI sharing and as an evaluative tool for assessing existing systems and platforms, this framework offers broad practical utility. While it offers a structured starting point, the framework does not claim to be exhaustive and should be interpreted as context-sensitive to the current evidence base (predominantly Western settings and heterogeneous study designs). Validation of the framework should proceed in stages (Delphi study, case-study application, and pilot testing). In ongoing work, we operationalize the framework in a habit-information PHI-sharing platform and aim to use A/B testing to evaluate selected factors on participation outcomes.

7 LIMITATIONS

While this systematic review offers valuable insights into the factors influencing participation in health data sharing, several limitations must be acknowledged. First, a formal quality appraisal of the included studies was not conducted, limiting the ability to assess the robustness of individual findings. Instead, generalizability was approximated based on study population characteristics. Furthermore, the studies varied significantly in terms of participant numbers which complicates the comparability and relative weight of their findings within the synthesis. Additionally, no exclusion criteria were applied based on methodological design, meaning that factors identified in literature reviews were given equal consideration to those derived from qualitative or quantitative empirical studies. Also, because some identified factors can be operationalized as persuasive design mechanisms, the framework should be applied under clear ethical safeguards (e.g., voluntariness, transparency, and avoidance of undue influence) rather than assuming that health-related objectives alone mitigate such concerns. Finally, although the qualitative synthesis was conducted collaboratively, interpretation bias cannot be entirely ruled out.

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